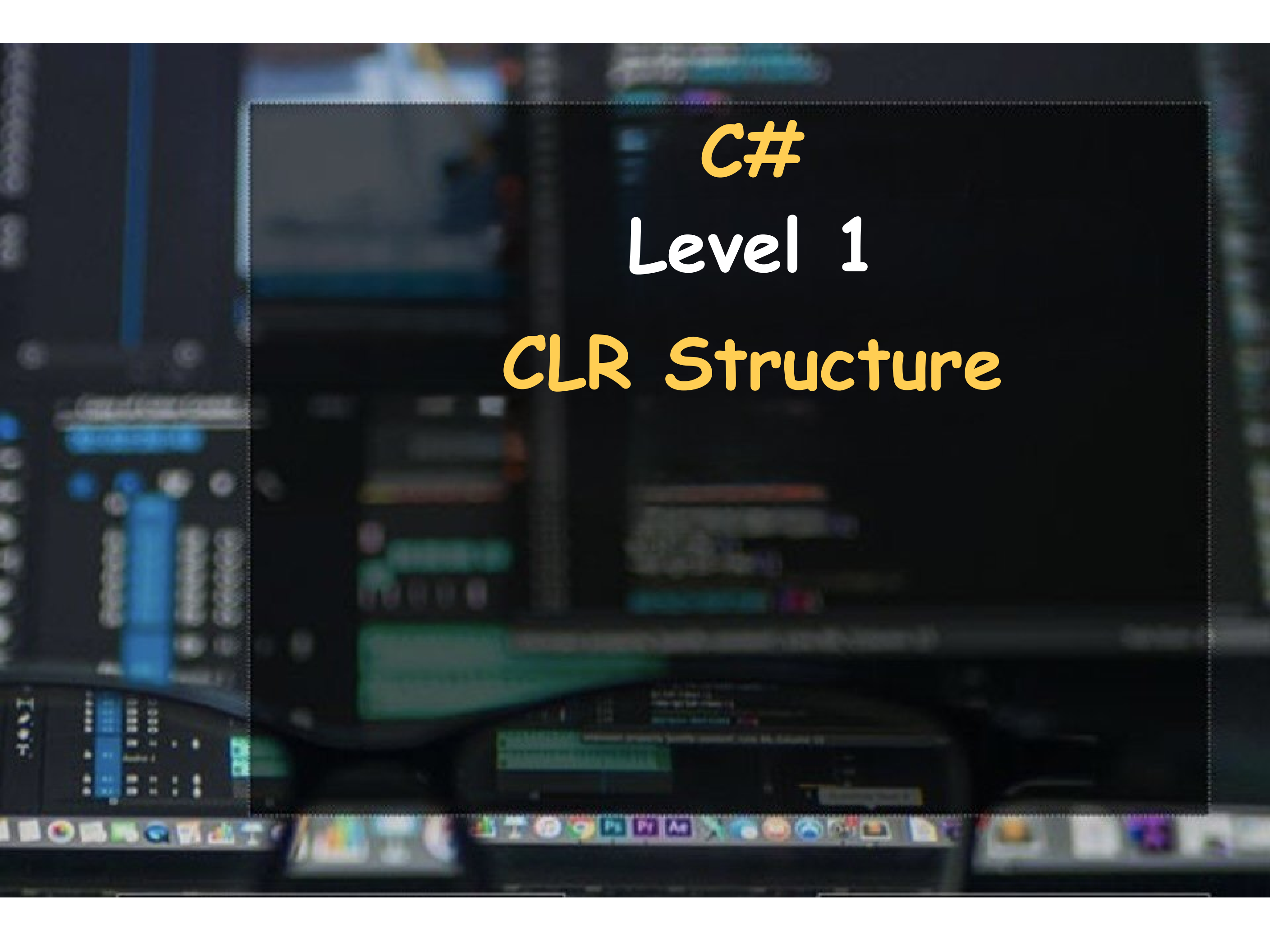




C#

Level 1

CLR Structure



C#

Level 1

CLR Structure

Definition

CLR Structure

 CLR contains multiple runtime components that work together to load, execute, and manage .NET applications.

 Important components include Class Loader, JIT, Garbage Collector, Type System, Exception Handling, and Thread Support.

Explanation

Class Loader

-  Loads required assemblies and types during execution.

JIT Compiler

-  Converts required IL into native machine code.

Garbage Collector

-  Manages managed memory and reclaims memory from unreachable objects.

Type Checker

-  Helps enforce rules of the .NET type system and type safety.

Exception Handler

-  Supports detecting, propagating, and handling runtime exceptions.

Thread Support

-  Supports threads, synchronization, and concurrent execution.

COM Marshaler

-  Helps managed code communicate with COM components.

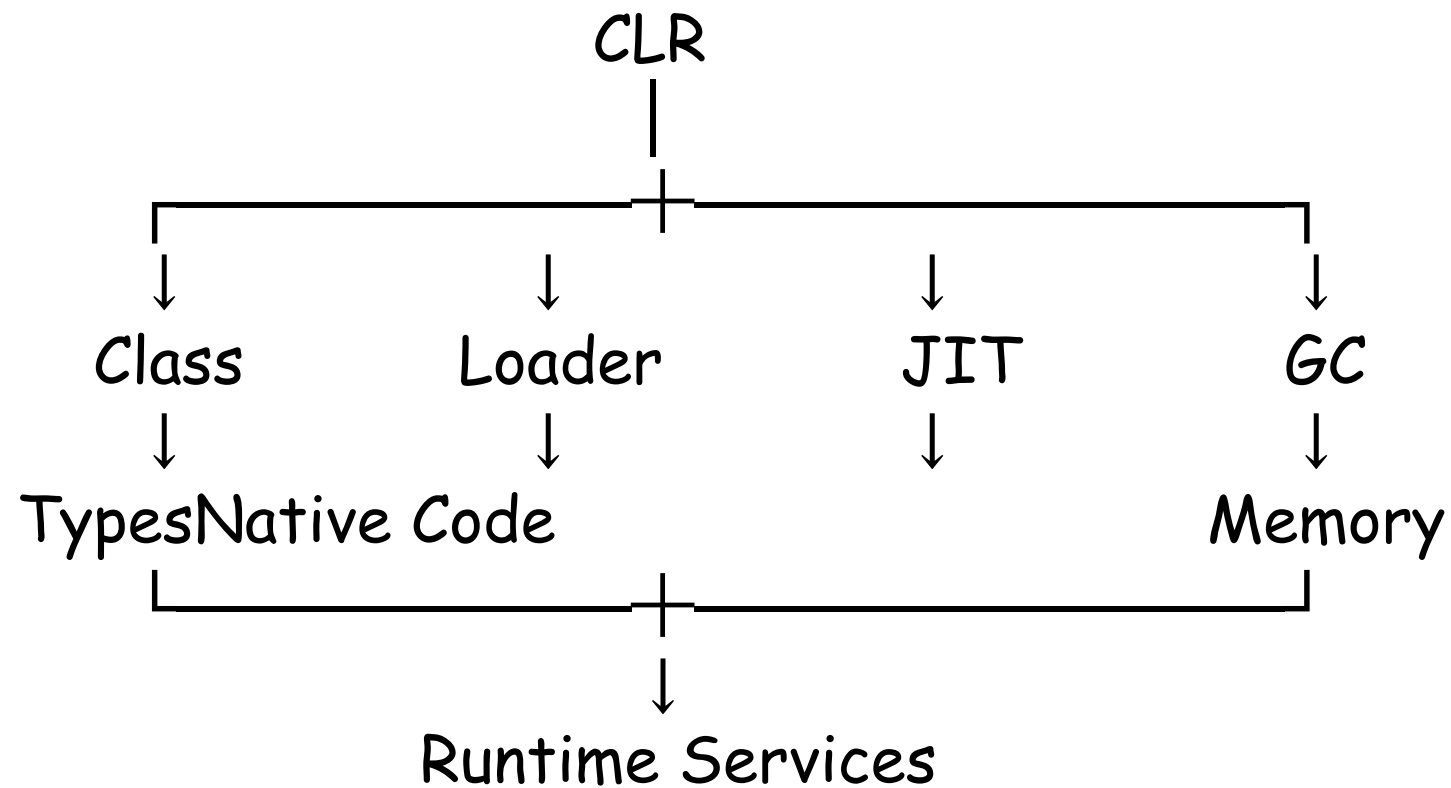
Debugging Support

-  Provides runtime support for debugging applications.

Procedures

1. CLR loads required assemblies and types.
2. JIT compiles required IL.
3. Runtime executes the application.
4. GC manages managed memory.
5. Other CLR services handle types, exceptions, threads, and interoperability.

💡 Example



➡ Result

The CLR coordinates different runtime services required to execute a .NET application.

Note

- ◆ This structure is based largely on classic .NET Framework terminology.
- ◆ Modern .NET keeps the core runtime concepts but has evolved internally.

⊘ Warning

- ◆ Do not memorize CLR as only JIT.
- ◆ JIT is one component of the runtime execution environment.



Comparison

Class Loader

- ✓ Loads assemblies and types.
- ✓ Makes required types available.

Garbage Collector

- ✓ Manages managed memory.
- ✓ Reclaims memory from unreachable objects.

JIT

- ✓ Converts IL to native code.
- ✓ Works during JIT-based execution.

Exception Handler

- ✓ Handles runtime exceptions.
- ✓ Supports exception propagation and handling.

✦✦ Summary

- ✓ CLR is made of multiple runtime services.
- ✓ Class Loader loads types and assemblies.
- ✓ JIT converts IL to native code.
- ✓ GC manages managed memory.
- ✓ Type checking supports type safety.
- ✓ Exception handling manages runtime errors.
- ✓ Thread support enables concurrent execution.

Thank you 😊